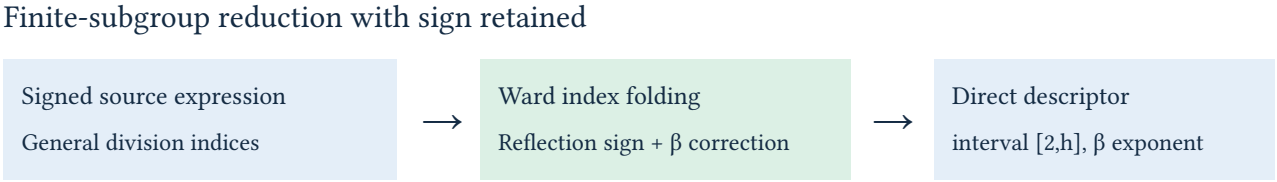
A compact projected-source interval

Phase-free finite-subgroup folding • T-1786 • 2026-11-07

The new constructor describes the projected source using one interval and a quasiperiod correction. It does not enumerate geometry or build an exponent map. For $h=(r-1)/2$, $S=h(h+1)(2h+1)/6$ and $\beta=\psi(r+1)^2\psi_2/\psi(r+2)$, the result is the projection of β to power $-S$ times the product of ψ_2 through ψ_h. Reduce the β exponent modulo the projection order q=2 or4.



r7: [2,3], β^2 r13: [2,6], β^1 r479: [2,239], β^0

Constructor stores five integers. Reference evaluation still visits each term.

Four software methods pass; 26 source comparisons; 1001 Ward-index checks.

PCC-20261107-01ee86 unsigned/unverified. No timing or faster-evaluation claim.

Why the reduction preserves the sign

Ward quasiperiodicity folds $n=kr+v$ to a signed half-range remainder. The implementation retains the reflection sign and β exponent; it does not simply replace n by its residue. The relation $\alpha^r=\beta^2$ eliminates α inside the projected image. The existing block pairs cover1 through h exactly once, yielding the direct interval constructor. No derivative seed, radical or field extension is needed by the new evaluator.

Native checks

Four selected methods pass;61 imported methods are filtered out. Folded and compact values match26 original projected-source cases. Single-term controls cover1001 indices, including multiples of the field characteristic. Flipping the sign fails all26 source cases; omitting the β correction fails at least one. The direct descriptor matches expanded folding in96 symbolic partition cases. Fourteen source/fixture hashes are pinned. The prior passing validation and exact source snapshot are retained; unused-code warnings remain in logs.

Exact remaining work

Compact construction is complete at this component's scope: five integer fields, no term vector. Its reference evaluator still visits h−1 division values and grows a cache. A faster interval-product evaluator preserving the odd-weight sign remains the next construction. The ambient divisor obstruction from the previous round remains valid; this is an equality on the finite subgroup. No production integration, experiment panel or speed measurement is claimed.

Evidence and sources

PCC-20261107-01ee86 is unsigned producer evidence; shared arithmetic means no independent verification. DERIVATION.md contains the formula, pairing proof, domains and replay details. Sources: Lauter–Stange2008, Theorem4 and section3, https://arxiv.org/pdf/0803.0728; actual source-divisor compiler, seed cancellation and original half-product construction. KB unavailable. The eprint mirror returned403; the primary arXiv paper was read.